

CIE-1 PYQ Bank: Model Answers

Handwritten Question Bank & Solutions

Syllabus Topics Covered:

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Unit 1 Examination Questions & Answers

UNIT 1

Verified model solutions for Unit 1 past exam questions.

Unit 1 5 Marks High Priority

CIE-1 2025SEE JAN 2026SEE 2025Faculty Priority Photo (Q1)

[Theory Link: Unit 1 § 1](#)

What is Artificial Intelligence? Explain the four major dimensions/approaches used to define AI systems (Acting Humanly, Thinking Humanly, Thinking Rationally, Acting Rationally).

Practice Problem:

Model Exam Answer

Solution:

1. Definition of Artificial Intelligence

Artificial Intelligence (AI) is the branch of computer science dedicated to creating software systems or hardware machines capable of performing tasks that typically require human intelligence, such as visual perception, speech recognition, reasoning, decision-making, and general problem-solving.

2. Four Approaches to AI (Russell & Norvig 2x2 Matrix)

Dimension	Human-Centered (Empirical Science)	Rationality-Centered (Ideal Standard)
Thinking	Thinking Humanly (Cognitive Science): Determining how the human mind thinks through psychological experiments and brain neuroimaging, then replicating that cognition in computer programs.	Thinking Rationally ("Laws of Thought"): Codifying right thinking through formal deductive logic (Aristotelian syllogisms). If premises are true, conclusions must be true.
Acting	Acting Humanly (The Turing Test): Creating machines that perform behaviors that would be considered intelligent if done by humans (Turing, 1950). Requires NLP, knowledge representation, automated reasoning, and machine learning.	Acting Rationally (Rational Agents): Designing computational agents that act to achieve the best outcome (or best expected outcome when uncertain). Modern AI focuses primarily on this approach!

Source: Russell & Norvig, AIMA 4th ed., Chapter 1.1, pp. 1-5

Unit 1 5 Marks High Priority

Faculty Priority Photo (Q2)CIE-1 2025

Theory Link: Unit 1 § 1

What are the types of AI? Explain both classification schemes (by capability and by functionality).

Practice Problem:

Model Exam Answer

Solution:

1. *Classification by Capability*

1. **Narrow / Weak AI (ANI):** AI trained and specialized for a single dedicated task. It excels in its domain but cannot generalize outside it.

Examples: Apple Siri, Google Translate, AlphaGo, Tesla Autopilot. (All existing AI today is Narrow AI).

2. **General / Strong AI (AGI):** Hypothetical AI possessing human-level cognitive capabilities across any intellectual task, capable of reasoning, abstract thinking, and transfer learning across domains.

3. **Super AI (ASI):** Theoretical future AI surpassing human intelligence across all disciplines, including scientific creativity, emotional intelligence, and social skills.

2. *Classification by Functionality (Hintze's Classification)*

- **Type 1: Reactive Machines:** Purely reactive; no memory or past experience used for decisions (e.g. IBM Deep Blue).
- **Type 2: Limited Memory:** Can look into the past for short intervals to make current decisions (e.g. self-driving cars monitoring recent vehicle velocities).
- **Type 3: Theory of Mind:** (Under research) AI that understands that humans and other entities have thoughts, emotions, and beliefs affecting their behavior.
- **Type 4: Self-Aware AI:** (Hypothetical) AI possessing conscious awareness of self and internal states.

Source: Russell & Norvig, Chapter 1; Faculty Slide Deck AI-Intelligent_Agents.pdf

<i>Parameter</i>	<i>Artificial Intelligence (AI)</i>	<i>Machine Learning (ML)</i>	<i>Deep Learning (DL)</i>
Scope	Broad umbrella concept of machines performing tasks requiring human intelligence.	Subfield of AI focusing on algorithms that learn patterns from data without explicit rule programming.	Subfield of ML based on multi-layered Artificial Neural Networks (ANNs).
Data Dependency	Works on rule engines, logic trees, or statistical models.	Performs well on structured/tabular medium datasets.	Requires massive volumes of data (millions of samples) to avoid overfitting.
Feature Engineering	Manually programmed rules and heuristics.	Requires manual feature extraction by domain experts.	Performs automated feature representation learning from raw data.
Hardware Needs	Standard CPUs.	CPUs / mid-range GPUs.	High-performance multi-GPU clusters and TPUs.
Examples	Expert systems, A* search, Minimax.	Random Forests, SVM, Logistic Regression, Linear Regression.	CNNs (computer vision), Transformers (LLMs, GPT-4), RNNs/LSTMs.

Unit 1 4 Marks High Priority

Faculty Priority Photo (Q4)

Theory Link: Unit 1 § 1

Discuss major real-world applications of Artificial Intelligence.

Practice Problem:

Model Exam Answer

Solution:

Key Application Domains of AI

1. **Healthcare & Medical Diagnosis:** Automated detection of tumors from MRI/CT scans (e.g. deep CNNs matching radiologist accuracy), predictive analytics for ICU patient deterioration, genomic drug discovery.
2. **Autonomous Systems & Transportation:** Self-driving vehicles (Tesla, Waymo) utilizing computer vision, LiDAR sensor fusion, and reinforcement learning for trajectory planning; automated drone pesticide spraying in precision agriculture.
3. **Natural Language Processing (NLP) & Assistants:** Large Language Models (ChatGPT, Claude), real-time multilingual translation (Google Translate), intelligent customer service agents.
4. **Finance & Banking:** High-frequency algorithmic trading, automated credit scoring, real-time credit card fraud detection via anomaly detection algorithms.
5. **Gaming & Robotics:** Superhuman game-playing agents (DeepMind AlphaZero, AlphaGo), automated warehouse logistics robots (Amazon Kiva robotic fulfillment).

Source: Russell & Norvig, Chapter 1.3, pp. 25–28

Unit 1 5 Marks High Priority

Faculty Priority Photo (Q5)SEE JAN 2026SEE 2025

Theory Link: Unit 1 § 2

What is an intelligent agent? Describe the structure of a simple intelligent agent interacting with its environment with a neat diagram.

Practice Problem:

Model Exam Answer

Solution:

1. Definition of an Agent

An **agent** is anything that can be viewed as perceiving its **environment** through **sensors** and acting upon that environment through **actuators**. Mathematically, an agent's behavior is described by an agent function $f : P^* \rightarrow A$, mapping any sequence of percepts P^* to an action A .

2. Agent-Environment Interaction Architecture



- **Percept:** The agent's perceptual inputs at any given instant.
- **Percept Sequence (P^*):** The complete history of everything the agent has ever perceived.

Source: Russell & Norvig, Chapter 2.1, pp. 36-39

<i>Agent Type</i>	<i>Sensors (Perception)</i>	<i>Actuators (Action)</i>
Human Agent	Eyes, ears, nose, tongue, skin receptors (touch, temperature).	Hands, legs, vocal cords, facial expressions.
Robotic Agent (e.g. Autonomous Car)	Cameras, LiDAR, RADAR, sonar, GPS, speedometer, odometer, gyroscopes.	Steering wheel motor, throttle/accelerator, brakes, display screen, turn signals.
Software Agent (e.g. Web Bot / Filter)	Keystrokes, received network packets, file contents, API requests.	Writing files, sending HTTP packets, displaying UI alerts, database queries.

Unit 1 5 Marks High Priority

Faculty Priority Photo (Q7)SEE 2025

[Theory Link: Unit 1 § 2](#)

What is a rational agent? Explain the four factors that determine rationality and differentiate rationality from omniscience.

Practice Problem:

Model Exam Answer

Solution:

1. Definition of Rationality

A **rational agent** is one that does the right thing. Specifically, for each possible percept sequence, a rational agent selects an action expected to maximize its performance measure, based on the evidence provided by the percept sequence and whatever built-in knowledge the agent possesses.

2. Four Factors Determining Rationality

1. **Performance Measure:** Defines the criterion of success for evaluating the sequence of environment states.
2. **Prior Knowledge:** The agent's initial background knowledge about its operational environment.
3. **Percept Sequence:** The historical sequence of observations received up to the current moment (P^*).
4. **Action Set:** The set of physical or logical actions that the agent is capable of executing.

3. Rationality $\neq$ Omniscience

- An **omniscient agent** knows the actual outcome of its actions and can act accordingly; but omniscience is impossible in reality.
- **Rationality** is about maximizing expected performance given the information available, whereas **omniscience** maximizes actual outcome. Rationality does not require clairvoyance!

Source: Russell & Norvig, Chapter 2.2, pp. 39-42

Unit 1 8 Marks High Priority

Faculty Priority Photo (Q8) CIE-1 2025SEE JAN 2026

[Theory Link: Unit 1 § 3](#)

Discuss the different types of intelligent agent architectures (Simple Reflex, Model-Based, Goal-Based, Utility-Based, and Learning Agents) with schematic diagrams.

Practice Problem:

Model Exam Answer

Solution:*Five Core Agent Architectures*

1. **Simple Reflex Agent:** Selects actions based solely on the current percept, ignoring history, using Condition–Action Rules (IF car-in-front-braking THEN brake). Fails in partially observable environments.
Pseudocode: `function SimpleReflex(percept): state = Interpret(percept); rule = RuleMatch(state, rules); return rule.Action.`
2. **Model-Based Reflex Agent:** Maintains an internal state tracking aspects of the environment that cannot be seen currently (updates internal model: 'How the world evolves' and 'How my actions affect the world'). Can handle partial observability.
3. **Goal-Based Agent:** Combines internal state information with a *goal* description to select actions that achieve desired future states. Uses search and planning.
4. **Utility-Based Agent:** Uses a *utility function* ($U : S \rightarrow \mathbb{R}$) mapping states to a real number expressing the degree of happiness or desirability. Enables trade-offs between conflicting goals under uncertainty.
5. **Learning Agent:** Divided into four conceptual components:
 - Learning Element: Responsible for making improvements.
 - Performance Element: Responsible for selecting external actions (equivalent to entire agent previously).
 - Critic: Evaluates agent behavior against an external performance standard.
 - Problem Generator: Suggests exploratory actions leading to new experiences.

Source: Russell & Norvig, Chapter 2.4, pp. 49–58

Parameter	Supervised Learning	Unsupervised Learning
Training Data	Trained on labeled data (pairs of input X and ground-truth target Y).	Trained on unlabeled data (only input features X , zero ground-truth labels).
Objective	Learn a mapping function $f(X) \rightarrow Y$ to accurately predict labels on unseen data.	Discover underlying hidden patterns, intrinsic groupings, or probability density of X .
Feedback Mechanism	Direct external supervisory error signal (loss = difference between prediction and actual label).	No external error feedback; model optimizes internal criteria (compactness, variance).
Major Sub-tasks	Classification (discrete labels) and Regression (continuous values).	Clustering, Dimensionality Reduction, Association Rule Mining, Density Estimation.
Algorithms	Linear Regression, Logistic Regression, Decision Trees, SVM, Neural Networks.	K-Means Clustering, Hierarchical Clustering, PCA, Autoencoders, Apriori.

<i>System</i>	<i>Performance Measure</i>	<i>Environment</i>	<i>Actuators</i>	<i>Sensors</i>
(i) Smart Agricultural Drone	Crop yield maximization, pesticide volume minimization, battery efficiency, zero collisions, flight safety.	Open farmland, crop rows, variable weather (wind, rain), trees, power lines.	Rotors/propellers, steering servos, pesticide spray nozzles, camera gimbal.	Multispectral RGB cameras, thermal sensors, LiDAR, GPS, altimeter, battery monitor.
(ii) Smart Traffic Control System	Vehicle throughput maximization, average wait time minimization, reduction in pedestrian delay, zero accidents.	City road intersection, multi-lane roads, vehicles, pedestrians, emergency vehicles.	Traffic light signals (Red/Yellow/Green timings), pedestrian crossing indicators, digital signage.	Inductive loop sensors, CCTV traffic cameras, radar vehicle detectors, pedestrian push buttons.
(iii) Medical Diagnosis System	Diagnostic accuracy, patient recovery rate, cost minimization, adverse reaction prevention.	Hospital, clinic, patient population, medical staff.	Display diagnostic report, treatment recommendations, prescription generation, referral alerts.	Keyboard input (symptoms, lab tests), medical imaging data feeds, electronic health records (EHR).

Unit 1 6 Marks High Priority

CIE-1 2025SEE 2025

Theory Link: Unit 1 § 3

Illustrate with a schematic diagram the working principle of a Simple Reflex Agent considering a two-state vacuum-cleaner environment.

Practice Problem:

Model Exam Answer

Solution:**1. Two-State Vacuum-Cleaner Environment**

Environment consists of two locations: Square *A* and Square *B*. Each square can be either Clean or Dirty. The agent can perceive its location and cleanliness: [*A*, Dirty], [*A*, Clean], [*B*, Dirty], [*B*, Clean].

2. Reflex Agent Action Rules

```
function VacuumReflexAgent([location, status]):  
    if status = 'Dirty':  
        return 'Suck'  
    else if location = 'A':  
        return 'Right'  
    else if location = 'B':  
        return 'Left'
```

3. Working Principle & Limitations

- **Direct Condition-Action:** The agent acts directly on the current percept without storing any memory of previous squares visited.
- **Limitation:** If the environment is partially observable (e.g. cleanliness sensor is disabled), the simple reflex agent cannot avoid infinite loops or determine whether cleaning was already completed.

Source: Russell & Norvig, Chapter 2.4, pp. 49-51

Unit 1 5 Marks Medium Priority

SEE JAN 2026

Theory Link: Unit 1 § 3

An online chess-playing system competes against human grandmasters. Analyze its task environment properties across the 7 standard environment dimensions.

Practice Problem:

Model Exam Answer

Solution:*Task Environment Characterization for Chess AI*

1. **Fully Observable vs Partially Observable: Fully Observable.** The complete board state and all piece positions are known to the agent at every turn.
2. **Single-Agent vs Multi-Agent: Multi-Agent (Competitive).** The agent plays against an opponent whose objective directly opposes the agent's objective.
3. **Deterministic vs Stochastic: Deterministic.** There is no randomness or dice; moving a piece to square X results with certainty in that piece occupying X .
4. **Episodic vs Sequential: Sequential.** Current moves fundamentally alter the future board position; actions have long-term consequences.
5. **Static vs Dynamic: Semi-Dynamic.** The physical board does not change while the agent thinks, but there is a time clock ticking down.
6. **Discrete vs Continuous: Discrete.** There are a finite number of distinct squares, distinct pieces, and legal turns.
7. **Known vs Unknown: Known.** The rules of chess are completely known and programmed into the game engine.

Source: Russell & Norvig, Chapter 2.3.2, pp. 44-47

Unit 2 Examination Questions & Answers

UNIT 2

Verified model solutions for Unit 2 past exam questions.

Unit 2 6 Marks High Priority

CIE-1 2025SEE JAN 2026

Theory Link: Unit 2 § 1

Explain how Bidirectional Search is efficient compared to BFS and DFS methods.
Show the BFS and DFS traversal for the following graph to reach goal state G:
Root A with children B, C; B has children D, E; C has child F; F has child G.

Practice Problem:

Model Exam Answer

Solution:**1. Efficiency of Bidirectional Search**

- Standard BFS searches forward from initial state to depth d with branching factor b , requiring time and space $O(b^d)$.
- **Bidirectional Search** runs two simultaneous searches: one forward from the initial state, and one backward from the goal, stopping when the two search frontiers meet in the middle!
- Complexity becomes $O(b^{d/2} + b^{d/2}) = O(2b^{d/2})$, which is an exponential reduction! E.g. for $b = 10$, $d = 6$, $b^d = 1,000,000$ nodes vs $2b^{d/2} = 2 \times 10^3 = 2,000$ nodes (a 500-fold reduction).

2. Traversal on Given Graph

Tree structure: $A \rightarrow \{B, C\}$, $B \rightarrow \{D, E\}$, $C \rightarrow \{F\}$, $F \rightarrow \{G\}$. Goal is G.

- **BFS (Breadth-First Traversal - Level by Level, FIFO Queue):**
 1. Visit Level 0: A
 2. Visit Level 1: B, C
 3. Visit Level 2: D, E, F
 4. Visit Level 3: G (Goal reached!)

• **Order of visited nodes:** $A \rightarrow B \rightarrow C \rightarrow D \rightarrow E \rightarrow F \rightarrow G$.
- **DFS (Depth-First Traversal - Deepest Node First, LIFO Stack):**
 Assuming left-to-right branch exploration:
 1. Explore left subtree: $A \rightarrow B \rightarrow D$ (dead end, backtrack to B)
 2. Explore next branch: E (dead end, backtrack to A)
 3. Explore right subtree: $C \rightarrow F \rightarrow G$ (Goal reached!)

• **Order of visited nodes:** $A \rightarrow B \rightarrow D \rightarrow E \rightarrow C \rightarrow F \rightarrow G$.

Unit 2 6 Marks High Priority

CIE-1 2025SEE 2025SEE JAN 2026

Theory Link: Unit 2 § 2

Solve the 8-puzzle problem using A^* search algorithm. Given Start State $[1, _, 3 / 4, 2, 6 / 7, 5, 8]$ and Goal State $[1, 2, 3 / 4, 5, 6 / 7, 8, _]$. Find the most cost-effective path (consider $g(n) = \text{depth}$ and $h(n) = \text{number of misplaced tiles}$).

Practice Problem:

Model Exam Answer

Solution:

1. Evaluation Function

$$f(n) = g(n) + h(n)$$

- $g(n)$: Path cost from root (depth level).
- $h(n)$: Number of misplaced tiles compared to Goal State $[1, 2, 3/4, 5, 6/7, 8, \text{blank}]$. (Blank tile is not counted).

2. Step-by-Step Search Tree Trace

Step 0: Start State (n_0)

1	_	3
4	2	6
7	5	8

- Compare with Goal $[1, 2, 3/4, 5, 6/7, 8, _]$: Tiles 2, 5, 8 are in wrong positions $\implies h(n_0) = 3$.
- $g(n_0) = 0 \implies f(n_0) = 0 + 3 = 3$.

Step 1: Expand Start Node (Valid moves for blank at top-middle: Left, Right, Down)

- **Move Left (Tile 1 moves Right):** $[_, 1, 3/4, 2, 6/7, 5, 8]$. Misplaced: 1, 2, 5, 8 ($h = 4$). $g = 1 \implies f = 1 + 4 = 5$.
- **Move Right (Tile 3 moves Left):** $[1, 3, _/4, 2, 6/7, 5, 8]$. Misplaced: 3, 2, 5, 8 ($h = 4$). $g = 1 \implies f = 1 + 4 = 5$.
- **Move Down (Tile 2 moves Up):**

1	2	3
4	_	6
7	5	8

Misplaced tiles: 5, 8 are wrong ($h = 2$). $g = 1 \implies f(n_1) = 1 + 2 = 3 \leftarrow$

Unit 2 6 Marks High Priority

CIE-1 2025SEE JAN 2026SEE 2025

Theory Link: Unit 2 § 4

Apply Minimax algorithm on the following game tree to compute the minimax value at each node: Max root A with Min children B and C. B has Max children D, E; C has Max children F, G. Leaf values: D has (3, 5); E has (6, 9); F has (1, 2); G has (9, -1).

Practice Problem:

Model Exam Answer

Solution:

1. Minimax Algorithm Principle

- **MAX node:** Chooses the maximum of its children: $V = \max(\text{child}_1, \text{child}_2)$.
- **MIN node:** Chooses the minimum of its children: $V = \min(\text{child}_1, \text{child}_2)$.

2. Bottom-Up Node Value Computation

Level 3 (Leaf Nodes):

- Children of D: 3, 5
- Children of E: 6, 9
- Children of F: 1, 2
- Children of G: 9, -1

Level 2 (MAX Nodes D, E, F, G):

- Node D (MAX): $\max(3, 5) = 5$
- Node E (MAX): $\max(6, 9) = 9$
- Node F (MAX): $\max(1, 2) = 2$
- Node G (MAX): $\max(9, -1) = 9$

Level 1 (MIN Nodes B, C):

- Node B (MIN): $\min(D, E) = \min(5, 9) = 5$
- Node C (MIN): $\min(F, G) = \min(2, 9) = 2$

Level 0 (Root MAX Node A):

- Node A (MAX): $\max(B, C) = \max(5, 2) = 5$

Optimal Move: Root A selects Left Branch (Node B) with Minimax Value = 5.

Source: Russell & Norvig, Chapter 5.2, pp. 163-166

Unit 2 6 Marks High Priority

CIE-1 2025

Theory Link: Unit 2 § 4

Explain Alpha-Beta Pruning. Define the terms α and β , state their roles, and explain how alpha-beta pruning improves efficiency over standard minimax.

Practice Problem:

Model Exam Answer

Solution:

1. Definition of Alpha and Beta

- α (**Alpha**): The value of the best (highest-value) choice found so far at any choice point along the path for **MAX**. Initialized to $-\infty$.
- β (**Beta**): The value of the best (lowest-value) choice found so far at any choice point along the path for **MIN**. Initialized to $+\infty$.

2. Pruning Condition (Alpha-Beta Cutoff)

At any node, if $\alpha \geq \beta$, the remaining children of that node do not need to be evaluated and are pruned (cut off). Because the parent already has a guaranteed outcome that is better than whatever this sub-branch could yield.

3. Efficiency Improvement over Standard Minimax

- Standard Minimax examines every leaf node in a tree with branching factor b and depth m : Time complexity $O(b^m)$.
- With **perfect move ordering** in Alpha-Beta pruning, the effective branching factor is reduced to $\sqrt{b}$. The time complexity drops to:

$$O(b^{m/2})$$

- This means Alpha-Beta pruning can search **twice as deep** in the same amount of CPU time as standard Minimax!

Source: Russell & Norvig, Chapter 5.3, pp. 167–172

Unit 2 6 Marks High Priority

CIE-1 2025

Theory Link: Unit 2 § 1

Why is uninformed search known as blind search? Explain Uniform Cost Search (UCS) with a suitable example.

Practice Problem:

Model Exam Answer

Solution:

1. Why Uninformed Search is 'Blind Search'

Uninformed search algorithms (BFS, DFS, Uniform Cost Search) have **no domain-specific knowledge** about how close a given state is to the goal. They can only generate successors and distinguish a goal state from a non-goal state; they have no clue whether one non-goal node is more promising than another.

2. Uniform Cost Search (UCS / Dijkstra's Algorithm)

- Instead of expanding the shallowest node (BFS), UCS expands the unvisited node with the **lowest path cost** $g(n)$ from the root.
- Implemented using a **priority queue** ordered by cumulative path cost $g(n)$.
- **Completeness & Optimality:** Guaranteed to find the optimal path if every step cost exceeds a positive constant $\epsilon > 0$.

3. Example: Route from S to G

Edges: $S \rightarrow A$ (cost 1), $S \rightarrow B$ (cost 5), $A \rightarrow G$ (cost 10), $B \rightarrow G$ (cost 2).

1. Init: Frontier = $\{[S, g = 0]\}$.

2. Expand S: Add A ($g = 1$), B ($g = 5$). Frontier = $\{[A, 1], [B, 5]\}$.

3. Expand A (min cost $g = 1$): Add G ($g = 1 + 10 = 11$). Frontier = $\{[B, 5], [G, 11]\}$.

4. Expand B (min cost $g = 5$): Add G ($g = 5 + 2 = 7$). Shorter path to G found! Frontier = $\{[G, 7]\}$.

5. Expand G ($g = 7$): Goal reached! Optimal path is $S \rightarrow B \rightarrow G$ with cost 7.

Source: Russell & Norvig, Chapter 3.4.2, pp. 83–86

Criterion	Breadth-First Search (BFS)	Depth-First Search (DFS)	Iterative Deepening Search (IDS)
Completeness	Yes (if branching factor b is finite).	No in infinite-depth spaces or with loops.	Yes (if b is finite).
Time Complexity	$O(b^d)$	$O(b^m)$ (where m is maximum depth; poor if $m \gg d$).	$O(b^d)$ (negligible overhead for repeating top levels).
Space Complexity	$O(b^d)$ (stores entire tree frontier; severe memory bottleneck!).	$O(b \cdot m)$ (linear space; highly memory efficient!).	$O(b \cdot d)$ (linear memory combined with BFS optimality!).
Optimality	Yes (for uniform step costs $g = 1$).	No (can find long path first).	Yes (for uniform step costs).

Unit 2 6 Marks High Priority

SEE 2024SEE 2025

Theory Link: Unit 2 § 2

What is an admissible heuristic? What is a consistent (monotonic) heuristic? Prove that A^* tree search is optimal if $h(n)$ is admissible.

Practice Problem:

Model Exam Answer

Solution:**1. Admissible Heuristic**

A heuristic function $h(n)$ is **admissible** if it never overestimates the true cost to reach the nearest goal state from node n :

$$0 \leq h(n) \leq h^*(n) \quad \forall n$$

where $h^*(n)$ is the true optimal path cost from n to goal. (E.g., Straight-line Euclidean distance is admissible for road navigation).

2. Consistent (Monotonic) Heuristic

A heuristic $h(n)$ is **consistent** if for every node n and every successor n' generated by action a with step cost $c(n, a, n')$:

$$h(n) \leq c(n, a, n') + h(n')$$

(Satisfies the triangle inequality. Every consistent heuristic is also admissible).

3. Proof Sketch: A^* is Optimal with Admissible Heuristic

- Suppose suboptimal goal G_2 is in the queue with $f(G_2) = g(G_2) > C^*$ (where C^* is optimal cost).
- Let n be an unexpanded node on the optimal path to true optimal goal G_1 .
- Since h is admissible: $f(n) = g(n) + h(n) \leq g(n) + h^*(n) = C^*$.
- Therefore: $f(n) \leq C^* < f(G_2)$.
- Since A^* always expands the node with minimum f in the priority queue, node n will always be selected before G_2 . Hence G_2 can never be selected for expansion, guaranteeing A^* returns the optimal goal!

Source: Russell & Norvig, Chapter 3.5.2, pp. 102–106

Unit 2 6 Marks Medium Priority

SEE 2024

Theory Link: Unit 2 § 3

Explain the Hill-Climbing search algorithm and discuss the three major problems that cause it to fail (Local Maxima, Ridges, and Plateaux).

Practice Problem:

Model Exam Answer

Solution:**1. Hill-Climbing Search Principle**

Hill-climbing (Greedy Local Search) is an iterative optimization algorithm that continuously moves in the direction of increasing elevation (value) to find the peak of a state landscape. It maintains only the current state and looks at its immediate neighbors.

2. Three Major Failure Modes

1. **Local Maxima:** A peak that is higher than each of its neighboring states, but lower than the global maximum. Since all neighboring moves look worse, the algorithm terminates prematurely.
2. **Ridges:** A sequence of local maxima connected together. The slope orientation causes navigation steps that zigzag without progressing toward the summit because single-direction orthogonal steps drop off on either side.
3. **Plateaux (Flat Local Maxima / Shoulders):** A flat area of the state space where all neighboring states have identical heuristic evaluation values. The search wanders aimlessly without guidance.

3. Remedial Variants

- **Random-restart hill climbing:** Conducts multiple hill-climbing runs starting from randomly chosen initial states.
- **Simulated Annealing:** Escapes local maxima by allowing occasional downhill moves with a probability that decreases over time (temperature cooling schedule).

Source: Russell & Norvig, Chapter 4.1.1, pp. 122-126

Unit 2 5 Marks Medium Priority

SEE 2025

[Theory Link: Unit 2 § 3](#)

Describe the Genetic Algorithm (GA) approach to local search. Explain its four core operators.

Practice Problem:

Model Exam Answer

Solution:**1. Genetic Algorithm (GA) Workflow**

Genetic algorithms are stochastic local search algorithms inspired by biological natural selection. A population of candidate solutions (encoded as strings over a finite alphabet, e.g. binary chromosomes) evolves over generations toward higher fitness.

2. Four Core Operators

- 1. Initial Population:** A collection of randomly generated individual states.
- 2. Fitness Function:** Evaluates each individual state and returns a numerical fitness score (e.g. number of non-attacking queen pairs in 8-queens). Higher fitness means higher probability of selection.
- 3. Selection & Crossover:** Pairs of individuals are selected for reproduction based on fitness (roulette wheel or tournament selection). A crossover point is chosen randomly, and substrings are swapped to produce offspring.
- 4. Mutation:** Each gene (character) in the offspring is flipped or modified with a small independent probability (mutation rate ≈ 0.01), introducing diversity and preventing premature convergence.

Source: Russell & Norvig, Chapter 4.1.4, pp. 130–133

Unit 2 5 Marks Medium Priority

SEE 2025

[Theory Link: Unit 2 § 4](#)

How can Artificial Intelligence solve the Tic-Tac-Toe problem? Explain with the help of a game tree representation.

Practice Problem:

Model Exam Answer

Solution:**1. Game Tree for Tic-Tac-Toe**

Tic-Tac-Toe is a 2-player, zero-sum, perfect-information, discrete, deterministic game played on a 3×3 grid.

- **Root:** Initial empty 3×3 board (9 possible moves for Player MAX / 'X').
- **Level 1:** 9 possible board states. Player MIN ('O') can choose from remaining 8 empty squares.
- **Terminal States:** A board where a player has 3 in a row (Utility: +1 if MAX wins, -1 if MIN wins, 0 if Draw), or all 9 squares are full.

2. Minimax Solution & Optimal Play

- Total state space is small: at most $9! = 362,880$ leaf nodes (reduced to 26,830 legal states via symmetry).
- Minimax searches the complete game tree to the terminal leaves and backs up optimal values.
- With optimal Minimax play, every game of Tic-Tac-Toe guarantees a **Draw (Utility = 0)**; MAX can never be beaten!

Source: Russell & Norvig, Chapter 5.1, pp. 161-164

Out of CIE-1 Scope Questions (10 Questions)

ARCHIVED

The following questions were identified in past papers but fall outside the official CIE-1 syllabus. They are preserved here to ensure zero questions are dropped.

Paper / Origin	Question Summary	Exclusion Rationale
SEE JAN 2026 Q5a	Write and interpret the propositional logic to represent knowledge for Wumpus world sentences (Square breezy, stench, glitter).	<u>Propositional logic and Wumpus world belong to Unit 3 (Knowledge Representation), out of CIE-1 scope.</u>
SEE JAN 2026 Q8a, SEE 2025 Q5a, CIE-2 2025 Q3b	Explain the process of Unification in First-Order Logic (FOL) with examples. Why is it essential for inference?	<u>First-Order Logic unification belongs to Unit 3/4.</u>
SEE JAN 2026 Q8b	Discuss the process of Knowledge Engineering in FOL. How can real-world knowledge be modelled?	<u>Knowledge Engineering in FOL belongs to Unit 4.</u>
SEE JAN 2026 Q9b, Q10a	Discuss the PDDL description of The simple spare tire problem and air cargo transportation.	<u>Planning and PDDL domain definition belongs to Unit 5.</u>
SEE JAN 2026 Q9a, Q10c, CIE-2 2025 Q1c	Can machines really think? Justify with 'Chinese Room' and 'Consciousness and qualia' arguments.	<u>Philosophy and Ethics of AI belongs to Unit 5.</u>
SEE 2025 Q5b, Q6a, CIE-2 2025 Q1b	Differentiate forward chaining and backward chaining in First-Order Logic.	<u>Inference chaining belongs to Unit 3.</u>
SEE 2025 Q6c, CIE-2 2025 Q2b	Explain the process that converts any first-order sentence into Conjunctive Normal Form (CNF).	<u>CNF conversion and resolution belong to Unit 3/4.</u>
SEE 2025 Q7a	Given full joint probability distribution for toothache/cavity/catch, calculate marginal and conditional probabilities.	<u>Probabilistic reasoning and joint distributions belong to Unit 4.</u>

Paper / Origin	Question Summary	Exclusion Rationale
SEE 2025 Q5c	Describe the properties of planning graphs which are monotonically increasing or decreasing.	<u>Planning graphs belong to Unit 5.</u>
CIE-2 2025 Q3c	Write PDDL description to solve the 'Blocks world problem'.	<u>Blocks world PDDL belongs to Unit 5.</u>